

Math 306 Group Quiz 4

Names: _____

1. Consider the sequence described by $r_0 = 0$, $r_1 = 1$ and $r_{n+1} = r_n + 2r_{n-1}$ for $n \geq 2$. This sequence is $0, 1, 1, 3, 5, 11, 21, \dots$

a. Let $T \in \mathcal{L}(\mathbb{C}^2)$ be $T(a, b) = (a + 2b, a)$. Use induction to show that $T^n(1, 0) = (r_{n+1}, r_n)$.

b. Find the eigenvalues/vectors for T . Observe that we have a basis of eigenvectors (an eigenbasis).

c. Express $(1, 0)$ in terms of this eigenbasis. Then simplify $T^n(1, 0)$ to find a formula for r_n .

2. Prove that if $T : V \rightarrow V$ is diagonalizable, then $V = \text{null } T \oplus \text{range } T$.